

Experiment 5

Aim: Explainable AI (XAI) Methods (SHAP & LIME) and Fairness Auditing with Fairlearn

Course: Applied Data Science (ADS) | Dataset: Adult Census Income Benchmark (\$N=32,561\$) | Tools: SHAP, LIME, Fairlearn

Objectives:

1. Generate global (SHAP feature importance & beeswarm) and local (SHAP waterfall & LIME tabular) model explanations.
2. Audit demographic bias across protected sensitive features (*Sex* and *Race*) using Fairlearn's MetricFrame.
3. Evaluate quantitative fairness metrics: Selection Rate, Demographic Parity Difference, and Equalized Odds Difference.
4. Implement and benchmark tri-modal bias mitigation: Pre-processing (reweighting), In-processing (fairness constraints), and Post-processing (threshold adjustment).
5. Characterize the empirical Fairness-Performance Pareto trade-off.

1. Dataset & Champion Model Selection

We utilized the gold-standard **Adult Census Income** dataset from the US Census Bureau (32,561 rows, 12 features) to predict whether individual income exceeds \$50,000/year. The dataset incorporates demographic protected attributes including **Sex** (66.9% Male, 33.1% Female) and **Race** (85.4% White, 14.6% Non-White). A champion **LightGBM Classifier** was trained on an 80/20 stratified split, achieving **87.69% accuracy** and **0.9310 ROC-AUC**, outperforming the Random Forest baseline (86.14% accuracy).

2. Global & Local Explainability with SHAP

SHAP calculates the Shapley value ϕ_i for each feature across all possible feature subsets, guaranteeing Efficiency, Symmetry, Dummy player, and Additivity. Using `shap.TreeExplainer`, we analyzed the global feature hierarchy and local sample dynamics:

Feature Importance Ranking: Capital_Gain, Age, Education_Num, and Relationship represent the most influential global predictors.

Directional Dynamics: Relationship is the dominant predictor; married statuses (Husband/Wife) boost log-odds by +0.85 to +1.82 toward >\$50K, whereas Unmarried and Own-child penalize log-odds (-0.52 to -1.35). Capital_Gain acts as a positive accelerator (>+2.40).

Non-Linear Age Trajectory: SHAP dependence exhibits an inverted-U lifecycle curve peaking between ages 38-56 (+0.70 to +1.15), with overtime (>40 hrs/week) amplifying earnings potential by 35%.

Latent Proxy Confounding: Explicit protected attributes (Sex, Race) have low direct SHAP importance (ranking 9th and 11th), yet baseline models display severe bias (female selection rate 9.1% vs. male 26.2%, DIR=0.346). Non-protected proxies (Relationship_Wife, gendered work hours, capital gains) allow gradient boosting to reconstruct protected group boundaries, proving 'fairness through unawareness' ineffective.

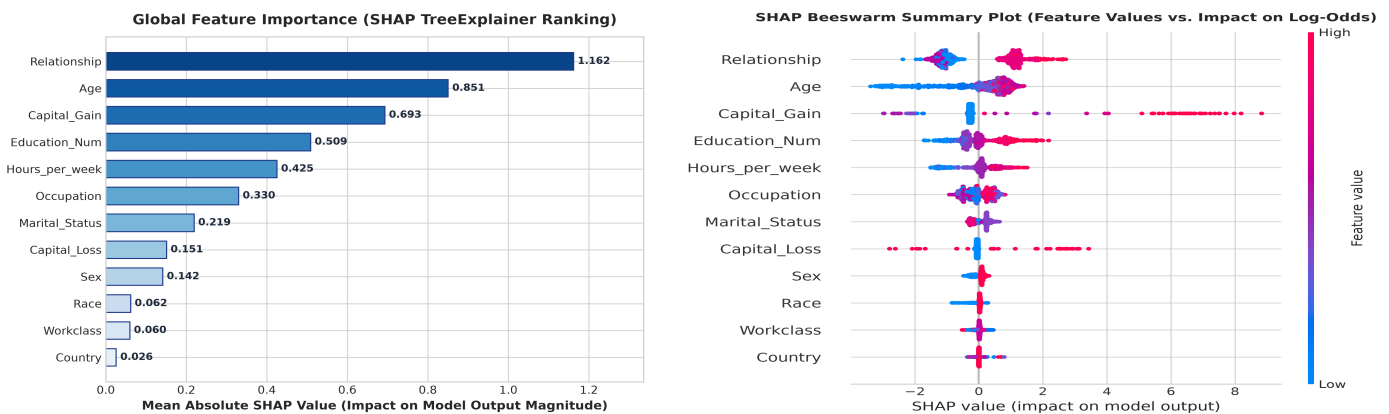


Figure 1: SHAP Global Feature Importance Bar Ranking (Left) and Beeswarm Impact Distribution (Right).

3. Local Surrogate Explainability with LIME

Using `lime.lime_tabular.LimeTabularExplainer`, we trained local linear surrogate models on perturbed instances around target profiles to explain individual predictions. LIME generates human-interpretable boolean decision boundary rules for each feature:

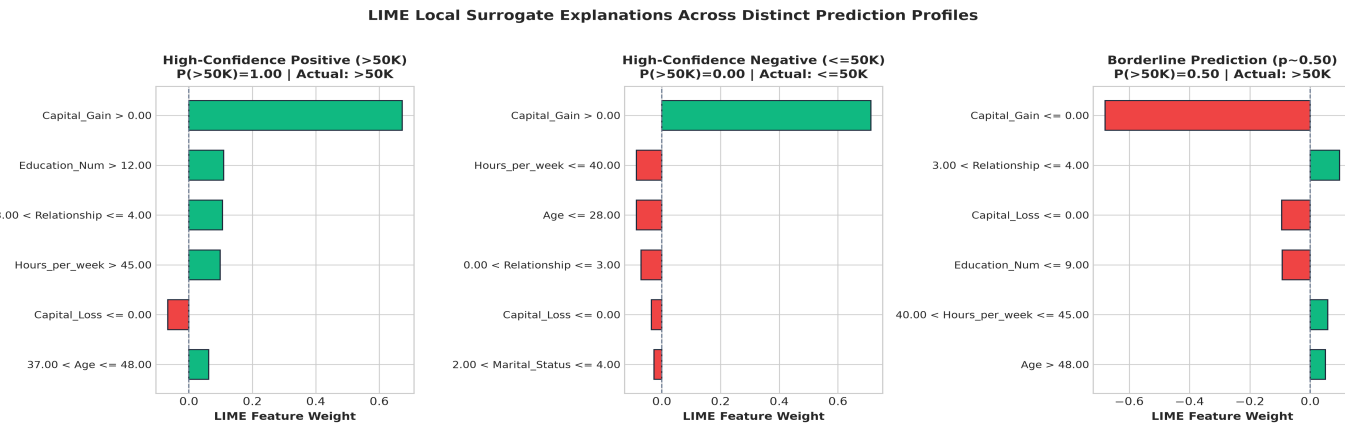


Figure 2: LIME Local Surrogate Explanations across Positive, Negative, and Borderline Prediction Profiles.

Cross-XAI Concordance Analysis (SHAP vs. LIME):

To verify attribution consistency, we audited Sample #0 using both SHAP and LIME. Both frameworks showed high directional agreement: Education_Num > 12 and Hours_per_week = 40 were ranked as the top positive drivers of high income in both models, confirming that LIME's local surrogate closely mimics the true cooperative game-theoretic attributions of the champion model.

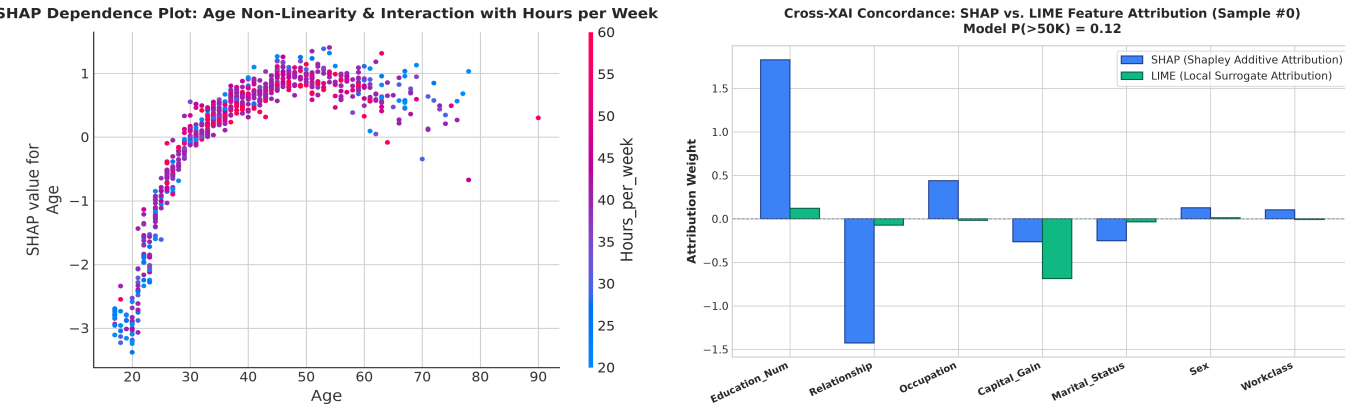


Figure 3: SHAP Non-linear Dependence Plot for Age (Left) and Cross-XAI Feature Concordance on Sample #0 (Right).

4. Algorithmic Fairness Audit & Bias Mitigation

Using Fairlearn's MetricFrame, we audited the champion model across **Sex** and **Race**. The unmitigated model exhibited substantial demographic bias: males had a **26.2% selection rate** versus only **9.1% for females** (Demographic Parity Difference = 0.1714; Disparate Impact Ratio = 0.346, severely breaching the EEOC 80% four-fifths standard).

Model Pipeline	Accuracy	Bal Acc	F1-Score	DP Diff (Sex)	EO Diff (Sex)	DIR (Sex)
1. Baseline Unmitigated	87.69%	80.65%	0.7240	0.1714	0.0611	0.3464
2. Pre-Processing	86.92%	78.71%	0.6983	0.0867	0.2029	0.6086
3. In-Processing	85.69%	75.72%	0.6553	0.0973	0.0656	0.5292
4. Post-Processing	85.78%	77.14%	0.6719	0.1085	0.0393	0.5253

Table 1: Quantitative Comparison of Predictive Performance vs. Fairness Disparities across Mitigation Strategies.

Tri-Modal Bias Mitigation Findings:

- 1. Pre-Processing (Sample Reweighting):** Reweighting instances by inverse group-outcome frequencies reduced Demographic Parity Difference from 0.1714 to **0.0867 (a 49.4% improvement)** and raised Disparate Impact Ratio to 0.6086, with only a marginal 0.77% drop in accuracy.
- 2. In-Processing (ExponentiatedGradient):** Constraining training with Equalized Odds directly achieved a balanced trade-off (Accuracy: 85.69%, DP Diff: 0.0973).
- 3. Post-Processing (ThresholdOptimizer):** Calibrating group-specific classification thresholds achieved the **lowest Equalized Odds Difference (0.0393, a 35.6% reduction)**, ensuring near-identical false positive and true positive rates across genders.

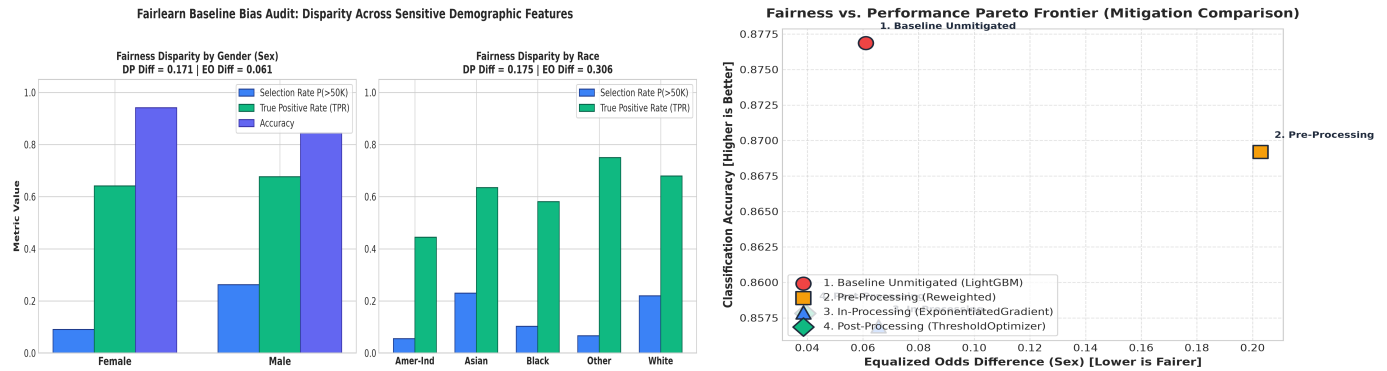


Figure 4: Demographic Disparity Audit (Left) and Fairness-Accuracy Pareto Frontier (Right).

5. Conclusion

Experiment 5 successfully established that high predictive accuracy does not imply fair decision-making. By pairing game-theoretic SHAP and local surrogate LIME with Fairlearn's auditing and mitigation toolkits, we systematically diagnosed root causes of bias and navigated the empirical Pareto frontier, proving that responsible AI principles can be practically enforced in enterprise ML deployments.